From YAML to Kernel Enforcement: Building a Sigma Rules Engine with eBPF LSM

How we compiled Sigma Rules into a stack-based expression evaluator running inside the Linux kernel.

Introduction

Sigma Rules look deceptively simple. A YAML file with some field names, a few string matches, and a condition like `selection and not filter`. Easy to read. Easy to write.

But try evaluating that logic inside an eBPF LSM hook — where you're on the critical path of different syscalls and need to make a block/allow decision as efficiently as possible — and things get complicated fast.

Consider what a real Sigma Rule condition can look like:

```
# See full rule below

condition: selection_passwd and (selection_suspicious_binaries or (selection_suspicious_uid and not
filter_whitelisted_binaries))
```

That's nested boolean logic with ANDs, ORs, NOTs, and parentheses. Each `selection_*` expands into multiple field comparisons — string contains, starts-with, numeric comparisons. The full expression tree can have dozens of nodes.

Now try running that inside the eBPF verifier's constraints: no recursion, no unbounded loops, 512-byte stack limit, and every memory access must be bounds-checked. Traditional expression evaluation techniques don't work here.

This post describes how we solved it in owLSM, an open-source stateful Sigma Rule Engine implemented via eBPF LSM. The pipeline compiles Sigma Rules into a kernel-evaluable format that runs in under a microsecond per syscall.

The Problem: A Concrete Example

Let's trace a specific rule through the entire pipeline. This rule blocks unauthorized write attempts to `/etc/passwd`:

```yaml
# More sigma metadata ...

description: "Block unauthorized /etc/passwd modifications"
action: "BLOCK_EVENT"
events:
    - WRITE
detection:

  selection_passwd:
      target.file.path|contains: "/etc/passwd"
  selection_suspicious_uid:
      process.euid|gt: 0
  selection_suspicious_binaries:
      process.file.path|endswith:
          - "ash"
          - "echo"
  filter_whitelisted_binaries:
      process.file.path|startswith:
          - "/usr/bin/apt"
          - "/usr/bin/yum"
          - "/usr/sbin/useradd"
  condition: selection_passwd and (selection_suspicious_binaries or (selection_suspicious_uid and not
filter_whitelisted_binaries))
```

The rule combines:

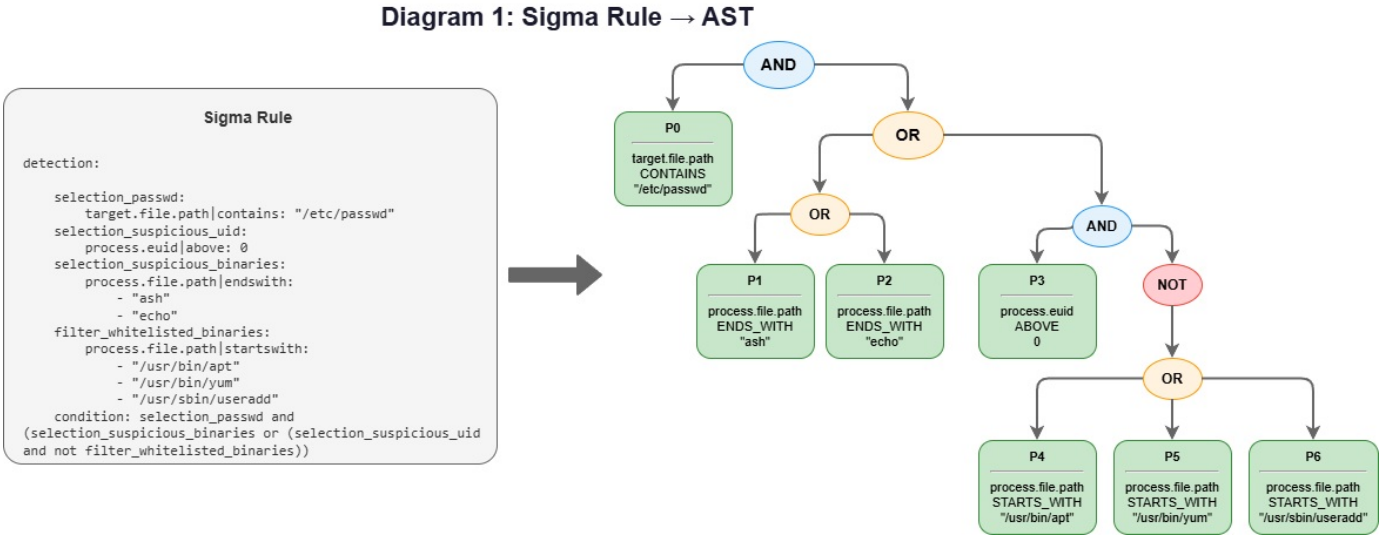
- 1 substring search (`contains`)
- 2 suffix matches (`endswith`)
- 3 prefix matches (`startswith`)
- 1 numeric comparison (`gt`)
- Nested ANDs, ORs, and a NOT

This needs to evaluate inside a kernel hook. Here's how we made it work.

Stage 1: Parsing — Sigma Rule to AST

AST Construction via pySigma

The detection logic is parsed into an Abstract Syntax Tree (AST) using [pySigma](), the official Sigma rule processing python library. We implemented a custom pySigma Backend that, extracts data from the AST and builds tables that will later be converted to bpf maps.



Each leaf node in the AST is an atomic **predicate** — a single field comparison:

Predicate	Field	Comparison	Value
P0	`target.file.path`	CONTAINS	“/etc/passwd”
P1	`process.file.path`	ENDS_WITH	“ash”
P2	`process.file.path`	ENDS_WITH	“echo”
P3	`process.euid`	ABOVE	0
P4	`process.file.path`	STARTS_WITH	“/usr/bin/apt”
P5	`process.file.path`	STARTS_WITH	“/usr/bin/yum”
P6	`process.file.path`	STARTS_WITH	“/usr/sbin/useradd”

Building the Lookup Tables

As pySigma walks the detection section, we build two deduplicated lookup tables:

`id_to_string`
The “Is Contains” flag marks strings needing KMP DFA pre-computation for substring search.

ID	Value	Is Contains
0	“/etc/passwd”	true
1	“ash”	false
2	“echo”	false
3	“/usr/bin/apt”	false
4	“/usr/bin/yum”	false
5	“/usr/sbin/useradd”	false

`id_to_predicate`
Field comparisons referencing strings/numbers

ID	Field	Type	Reference
0	target.file.path	CONTAINS	str: 0
1	process.file.path	ENDS_WITH	str: 1
2	process.file.path	ENDS_WITH	str: 2
3	process.euid	ABOVE	num: 0
4	process.file.path	STARTS_WITH	str: 3
5	process.file.path	STARTS_WITH	str: 4
6	process.file.path	STARTS_WITH	str: 5

Predicates, strings and IP addresses are **deduplicated** across rules. If 10 different rules all check `target.file.path contains "/etc/passwd"`, that predicate and “/etc/passwd” are stored once and referenced by index everywhere. This deduplication is critical — it reduces kernel memory and enables predicate result caching.

Stage 2: Transformation — AST to Postfix (Reverse Polish Notation)

We need to evaluate the AST inside eBPF, but:

- **No recursion** — the verifier forbids it
- **No call stack** — we can't traverse a tree with function calls
- **512-byte stack limit** — we can't store arbitrary nested structures. verifier limitation

The solution: **convert the AST to postfix notation** (Reverse Polish Notation). Postfix expressions can be evaluated with a single linear pass using a fixed-size stack — no recursion, no operator precedence logic, no parentheses needed at evaluation time.

We use **infix-to-postfix conversion** via tree traversal. The specific traversal order is determined by the pySigma backend implementation.

Our example rule produces this postfix sequence:

```
P0 P1 P2 P3 P4 P5 P6 OR OR NOT AND OR OR AND
```

Tokenization

Each element in the postfix sequence is converted to a **token** — a small struct containing:

- `operator_type`: PREDICATE, AND, OR, or NOT
- `predicate_idx`: (only for PREDICATE tokens) index into the predicate table

Serialization to JSON

The lookup tables and token arrays are serialized into JSON:


```json
{
  "id_to_string": {
    "0": { "value": "/etc/passwd", "is_contains": true },
    "1": { "value": "ash", "is_contains": false },
    "2": { "value": "echo", "is_contains": false },
    "3": { "value": "/usr/bin/apt", "is_contains": false },
    "4": { "value": "/usr/bin/yum", "is_contains": false },
    "5": { "value": "/usr/sbin/useradd", "is_contains": false }
  },
  "id_to_predicate": {
    "0": { "field": "target.file.path", "comparison_type": "contains", "string_idx": 0 },
    "1": { "field": "process.file.path", "comparison_type": "endswith", "string_idx": 1 },
    "2": { "field": "process.file.path", "comparison_type": "endswith", "string_idx": 2 },
    "3": { "field": "process.euid", "comparison_type": "above", "numerical_value": 0 },
    "4": { "field": "process.file.path", "comparison_type": "startswith", "string_idx": 3 },
    "5": { "field": "process.file.path", "comparison_type": "startswith", "string_idx": 4 },
    "6": { "field": "process.file.path", "comparison_type": "startswith", "string_idx": 5 }
  },
  "id_to_ip": {},
  "rules": [
    {
      "id": 42,
      "description": "Block unauthorized /etc/passwd modifications",
      "action": "BLOCK_EVENT",
      "applied_events": ["WRITE"],
      "tokens": [
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 0 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 1 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 2 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 3 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 4 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 5 },
        { "operator_type": "OPERATOR_PREDICATE", "predicate_idx": 6 },
        { "operator_type": "OPERATOR_OR" },
        { "operator_type": "OPERATOR_OR" },
        { "operator_type": "OPERATOR_NOT" },
        { "operator_type": "OPERATOR_AND" },
        { "operator_type": "OPERATOR_OR" },
        { "operator_type": "OPERATOR_OR" },
        { "operator_type": "OPERATOR_AND" }
      ]
    }
  ]
}
```


This JSON becomes part of the configuration passed to the userspace component at startup.
Now we will see how the userspace component deserializes the json and uses it to populate bpf maps.

Stage 3: Pre-computation — Userspace Preparation

Before anything reaches the kernel, the userspace component loads the JSON and pre-computes everything it can.

The KMP DFA for Substring Search

The `CONTAINS` comparison needs substring search.
The naive nested loop O(n×m) approach is too slow for inline syscall evaluation and will not pass the verifier due "instructions explosion"
KMP (Knuth-Morris-Pratt) gives us O(n), but standard KMP uses a failure function with conditional branches — problematic in eBPF where we want predictable execution.

We build a **complete DFA transition table** — a 2D array where `dfa[state][character]` gives the next state directly. No conditionals, no failure-function chasing.

For the pattern "/etc/passwd", the DFA has states 0 through 11 (pattern length), and each state has 256 entries (one per possible byte value).
Reaching the final state means the pattern was found.

The trade-off is memory: each DFA is a fixed `256 × 256` byte array. This is a significant memory cost, but the guarantee of worst-case O(n) substring functionality which is verifier friendly, is worth it for inline syscall evaluation.

Event-Type Rule Maps

Rules are stored in **per-event-type bpf arrays**. A rule with `events: [WRITE]` is only inserted into `write_rules`.
When a WRITE syscall fires, the kernel iterates *only* over `write_rules` — never touching READ, EXEC, or NETWORK rules.

A rule can specify multiple event types (e.g., `events: [CHMOD, CHOWN, READ, WRITE]`) and will be inserted into all relevant maps. We won't expand on multi-event rules in this post.

BPF Map Population

After pre-computation, userspace populates the BPF maps:

- `predicates_map` — Global predicate table. equivalent to id_to_predicate.
- `rules_strings_map` — String values with length and DFA index. equivalent to id_to_string.
- `idx_to_DFA_map` — Pre-computed KMP DFAs for `contains` strings
- `rules_ips_map` — IP addresses in binary form with CIDR masks. Not talked about in this post.
- `{event}_rules` — Per-event-type rule arrays
- `predicates_results_cache` — Predicate result cache (per-CPU)

After populating the maps, the userspace attaches the eBPF probes.

Stage 4: Kernel Evaluation — The Stack Machine

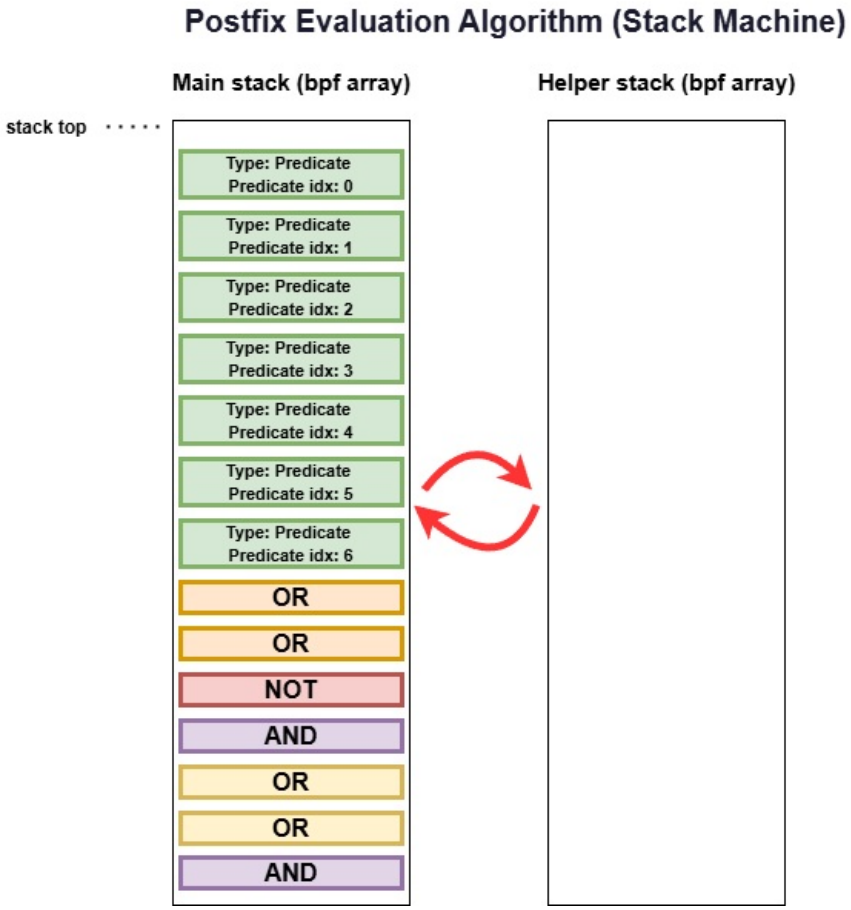
When a WRITE syscall fires, the eBPF hook runs.

Event Population

First, the hook populates an event struct with all relevant context:
`target.file.path` = “/etc/passwd”, `process.euid` = 1000 and dozens of other event fields.
The event timestamp is also captured here — this becomes important for `predicates_results_cache`.

Stack-Based Postfix Evaluation

We use the **reverse Polish notation evaluation algorithm**. This is the standard algorithm for evaluating postfix expressions — it’s O(n) in the number of tokens and requires only 2 small fixed-size stacks.
We implemnted the stacks using bpf arrays. Implementing `top`, `pop`, `push`, `empty` functions.



After processing all tokens, the stack contains exactly one value: the final result.

Predicate Lookup and Evaluation

When we need to evaluate a PREDICATE token, we:

1. Read `predicate_idx` from the token
2. Check if its in the `predicates_results_cache`
3. Look up the predicate struct from `predicates_map[predicate_idx]`
4. Dispatch based on comparison type: string, numeric or IP comparison.
5. store the result in `predicates_results_cache`

Predicate Result Caching

The same predicate often appears in multiple rules.
Without caching, we’d evaluate `target.file.path contains "/etc/passwd"` once per rule — potentially dozens of times.
We maintain a `predicates_results_cache` per-CPU BPF hash map.
But how do we invalidate it between events without expensive cleanup?

Timestamp-based invalidation: Each cache entry stores the event timestamp alongside the result.
When checking the cache:

- If the stored timestamp matches the current event's timestamp → cache hit, return stored result
- If timestamps differ → cache miss, evaluate and store with new timestamp

This works because:

1. The event timestamp is captured during event population (Stage 4)
2. The cache is a **per-CPU map** — no cross-CPU interference
3. eBPF programs are **non-preemptible** — once an event starts processing on a CPU, it runs to completion before any other event on that CPU

This guarantees that within a single event, all cache lookups see consistent timestamps. When the next event arrives, it has a higher timestamp, automatically invalidating all previous entries without any explicit cleanup.

```
cached_result = get_cached_result(predicate_idx, current_event_timestamp);
if (cached_result != UNKNOWN)
    return cached_result;

result = evaluate_predicate(predicate_idx, event);
cache_result(predicate_idx, result, current_event_timestamp);
return result;
```

First Match Wins

Rules are stored sorted by ID (lowest first).
When a rule matches, evaluation stops immediately — Rule 1 is evaluated before Rule 100.
This differs from most Sigma Rules Engines that always evaluate all rules and aggregate results. For inline enforcement where microseconds matter, early exit is critical.

What We Left Out

This post focused on the core algorithm chain.
We explicitly didn't cover many things like: fieldref, keywords, IP matching, and many more cool features.
Let me know if you'd like to hear about how we solved these.